Cape Peninsula
University of Technology

GA2 REPORT: "ARDUINO-CONTROLLED FREQUENCY RESPONSE MEASUREMENT OF AN ACTIVE LOW-PASS FILTER"

Nnete Thapedi Teise

Module: MCP261S
Group B
Allocation: R = 100 kΩ, C = 10 nF, Profile C

Abstract

This report looks at the frequency response of an active first-order low-pass filter simulated in SimulIDE. The filter uses a 100 kΩ resistor, a 10 nF capacitor and an op-amp wired as a unity-gain buffer. Five sine-wave sources were switched in one at a time by Arduino-controlled relays, at 0.25, 0.50, 1.00, 1.50 and 2.50 times the calculated cutoff frequency of 159.155 Hz (Profile C). Gain was worked out from oscilloscope readings of the input and output voltages and compared with the first-order model $|H| = 1/\sqrt{(1 + (f/f_c)^2)}$.

Measured gain dropped from 0.968 (−0.28 dB) at 39.78 Hz to 0.372 (−8.59 dB) at 397.88 Hz. At the cutoff frequency it was 0.708 (−3.00 dB), against a theoretical 0.707 (−3.01 dB). No result differed from theory by more than 0.27%. The results back up the first-order, unity-passband-gain model for the simulated circuit. They say nothing about physical hardware, and phase was not measured.

Keywords: active low-pass filter; RC filter; op-amp voltage follower; frequency response; cutoff frequency; Arduino; relay; SimulIDE.

1. Mathematical and Scientific Terms

This section sets out the terms and quantities used in the rest of the report (Alexander and Sadiku, 2017; Hambley, 2018). Table 1 gives each one with its symbol, unit and meaning in this circuit.

Table 1: Terms, symbols and units used in this report.

Term	Symbol and unit	Definition
Resistance	R (Ω)	The opposition to current flow. In this circuit R = 100 kΩ.
Capacitance	C (F)	The ability of a component to store charge. In this circuit C = 10 nF = 10×10^{-9} F.
Capacitive reactance	X_C (Ω)	The frequency-dependent opposition of a capacitor to alternating current, $X_C = 1/(2\pi fC)$.
Time constant	τ (s)	The product of resistance and capacitance, $\tau = RC$. It describes how quickly the capacitor charges and discharges.
Cutoff frequency	f_c (Hz)	The frequency at which the output amplitude falls to $1/\sqrt{2}$ (about 70.7%) of the passband level, a drop of 3 dB. For a first-order RC filter, $f_c = 1/(2\pi RC)$.
Input voltage	V_{in} (V)	The sinusoidal voltage measured at the filter input, after the relay selector.
Output voltage	V_{out} (V)	The voltage measured at the output of the op-amp.
Gain	A_v (no unit)	The ratio of output amplitude to input amplitude, $A_v = V_{out}/V_{in}$. A gain below 1 means the signal is attenuated.
Attenuation	–	A reduction in signal amplitude, which means the gain is less than 1 (negative in dB) and the output is smaller than the input.
Passband gain	K (no unit)	The gain at frequencies well below f_c. For a unity-gain buffer K = 1 (0 dB).

Term	Symbol and unit	Definition
Decibel	dB	A logarithmic unit for gain, $A_v(dB) = 20 \log_{10}(V_{out}/V_{in})$. A gain of 1 is 0 dB and a gain of 0.707 is about −3 dB.
Frequency response	H(f)	The way gain and phase change with frequency.
Phase shift	ϕ (°)	The angle by which the output sinusoid leads or lags the input sinusoid.
Active low-pass filter	–	A filter that passes low frequencies, attenuates high frequencies and includes an active device. Here the active device is an op-amp connected as a unity-gain buffer.
Relay selector	–	A set of relays that connects one source at a time to the filter input. Each relay coil is driven by a transistor controlled by an Arduino pin.
Amplitude convention	V_p (V)	All amplitudes in this report are peak values (zero-to-peak), not peak-to-peak values.

2. Theory in Engineering Problems

2.1 Overall circuit

The circuit has five SimulIDE Wave Generators, each set to a different frequency (SimulIDE, n.d.). Every generator has its own relay contact, and the five relay outputs meet at one node, V_{in}. That node feeds a series resistor R, and a capacitor C runs from the point after R to ground. The RC junction goes to the non-inverting (+) input of an op-amp, and the op-amp output is tied back to its inverting (−) input, which makes it a unity-gain buffer. An oscilloscope reads V_{in} on Ch1 and V_{out} on Ch2. Figure 1 (Section 5.1) shows the whole circuit.

2.2 Low-pass behaviour

A first-order RC low-pass filter works because the reactance of a capacitor depends on frequency, $X_C = 1/(2\pi fC)$ (Alexander and Sadiku, 2017). At low frequencies X_C is large compared with R, so very little current goes into the capacitor and the signal gets through almost unchanged. As the frequency goes up, X_C gets smaller, more of the signal is shunted to ground and the output shrinks. When $f = f_c$, R and X_C have the same magnitude. The output is then $1/\sqrt{2} \approx 0.707$ of the input and lags it by 45° (Hambley, 2018). With the allocated components, X_C at f_c is $1/(2\pi \times 159.155 \times 10 \times 10^{-9}) \approx 100$ kΩ, the same as R, so the numbers agree.

2.3 Why an op-amp buffer is used

The op-amp is wired as a voltage follower: the RC junction drives the non-inverting input and the output goes straight back to the inverting input. The output therefore copies the junction voltage without inverting it, so the passband gain is K = 1 (0 dB). The follower also has a very high input impedance and a low output impedance (Sedra et al., 2020; Franco, 2015). Whatever is connected to V_{out}, such as an oscilloscope probe, takes almost no current from the RC node, so it cannot move the cutoff frequency. Without the buffer, a load of around 100 kΩ would form a divider with R and change the response

being measured. Adding an active device like this is what makes the filter active rather than passive (Mancini, 2002).

2.4 Op-amp supply arrangement

The input signal swings both above and below 0 V, so the op-amp needs a negative supply as well as a positive one. A dual supply with its midpoint at ground is needed, and each rail has to be larger than the 2.50 V input peak or the output will be limited (Sedra et al., 2020). An earlier test with too little negative supply clipped the negative half of V_{out} at 0 V. Once a dual supply was added, V_{out} was a full sine wave, which shows how much the supply arrangement affects the measurement.

2.5 Arduino-controlled source selection

An Arduino Uno output pin has a recommended limit of 20 mA (Arduino, n.d.a), but the relay coil needs about 50 mA (Section 2.7), so a pin cannot drive the coil directly. Instead, each pin (D2 to D6) goes through a 1 kΩ base resistor to an NPN transistor. When the pin is HIGH (5 V) and the base–emitter voltage is about 0.7 V, the base current is roughly (5 − 0.7)/1000 ≈ 4.3 mA, well inside the pin limit. The transistor switches on, current flows from +5 V through the coil and the relay contact closes. Sources F1 to F5 are assigned to pins D2 to D6 respectively, and each channel is active HIGH (relay closed) and LOW when inactive.

A flyback diode sits across each coil. When the transistor turns off, the coil inductance keeps the current flowing, and the diode gives that current somewhere to go. Without it, the voltage spike could damage the transistor (Horowitz and Hill, 2015). The program closes only one relay at a time and switches every relay off for a moment between channels (break-before-make), so two sources are never joined together. The signal only passes through the relay contacts, and the Arduino does not generate any sine wave.

2.6 Program timing

The code times everything with millis() instead of a long blocking delay (Arduino, n.d.b), so the program is never stuck waiting. Each source stays connected for 10 s (the dwell time), followed by a 0.3 s gap with all relays off. The sequence runs F1, F2, F3, F4, F5 and then repeats, so one full cycle takes 5 × (10 + 0.3) = 51.5 s. The selected source and its frequency are printed to the Serial Monitor.

2.7 Relay settings

The relay is a single-pole, single-throw (SPST) normally-open type. Its coil resistance is 100 Ω, coil inductance 100 mH, turn-on current (I_{On}) 20 mA and turn-off current (I_{Off}) 10 mA. With 5 V across the coil the current is about 5 V / 100 Ω = 50 mA, comfortably above the 20 mA needed to close the relay.

3. Mathematical Methods

The same steps were applied to every test frequency. The equations come from standard circuit-analysis texts (Alexander and Sadiku, 2017; Sedra et al., 2020).

3.1 Allocated values

The allocation file gives each student a resistance (R_ohm), a capacitance (C_nF), a profile from A to E and five multipliers for F1 to F5. For this report the allocation is R = 100 kΩ, C = 10 nF and Profile C, with multipliers 0.25, 0.50, 1.00, 1.50 and 2.50. The test frequencies in hertz are then calculated from the cutoff frequency.

3.2 Time constant and cutoff frequency

$$\tau = RC \tag{1}$$

$$f_c = 1/(2\pi RC) \tag{2}$$

3.3 Test frequencies

Each multiplier m is applied to the cutoff frequency.

$$f = m \times f_c \tag{3}$$

3.4 Measured gain

Gain is the output amplitude divided by the input amplitude. V_{in} is measured after the relay selector and V_{out} at the filter output, and both use the same amplitude convention (peak).

$$A_v = V_{out}/V_{in} \tag{4}$$

$$A_v(\mathrm{dB}) = 20\log_{10}(V_{out}/V_{in}) \tag{5}$$

3.5 Theoretical gain and phase

For a first-order low-pass filter with passband gain K, the magnitude and phase of the transfer function are given by Equations 6 and 7. Here K = 1.

$$|H| = K / \sqrt{(1 + (f/f_c)^2)} \tag{6}$$

$$\varphi = -\arctan(f/f_c) \tag{7}$$

The theoretical gain in dB is $20\log_{10}|H|$. At $f = f_c$ the phase is −45°, which is a time lag of $\Delta t = 1/(8f_c) \approx 0.785$ ms.

3.6 Expected output amplitude

$$V_{out,exp} = V_{in} \times |H| \tag{8}$$

3.7 Percentage difference

The theoretical gain is the reference value. Intermediate values were kept to at least five significant figures where possible, and final gains are quoted to three decimal places and dB values to two.

$$\% \text{ difference} = (A_{v,\text{measured}} - |H|) \,/\, |H| \times 100 \tag{9}$$

4. Calculations

4.1 Time constant and cutoff frequency

Using Equation 1, with R = 100 kΩ and C = 10 nF:

$$\tau = 100 \times 10^{3}\ \Omega \times 10 \times 10^{-9}\ \text{F} = 1.000 \times 10^{-3}\ \text{s} = 1\ \text{ms} \tag{10}$$

Using Equation 2:

$$f_c = 1/(2\pi \times 1.000 \times 10^{-3}\ \text{s}) = 159.155\ \text{Hz} \tag{11}$$

4.2 Test frequencies

Applying Equation 3 to each multiplier gives the test frequencies in Table 2. The frequencies typed into the generators are quoted to two decimal places in Table 5.

Table 2: Test frequencies, $f = m \times f_c$.

Source	Multiplier m	Calculation	Frequency (Hz)
F1	0.25	0.25 × 159.1549	39.789
F2	0.50	0.50 × 159.1549	79.577
F3	1.00	1.00 × 159.1549	159.155
F4	1.50	1.50 × 159.1549	238.732
F5	2.50	2.50 × 159.1549	397.887

4.3 Theoretical gain and phase

Equations 6 and 7 were evaluated with K = 1 at the exact frequencies $f = m \times f_c$, so that $f/f_c = m$. For F1 (m = 0.25):

$$|H| = 1/\sqrt{(1 + 0.25^{2})} = 1/\sqrt{1.0625} = 0.97014 \approx 0.970 \tag{12}$$

$$20 \log_{10}(0.970) = -0.26\ \text{dB} \tag{13}$$

$$V_{\text{out,exp}} = 0.970 \times 2.50\ \text{V} = 2.425\ \text{V} \tag{14}$$

The same procedure for F2 to F5 gives the values in Table 3, and the handwritten workings are in Appendix A.

Table 3: Theoretical gain and phase for K = 1.

Source	f/f_c	$1 + (f/f_c)^2$	$\lvert H \rvert$	$20 \log_{10}\lvert H \rvert$ (dB)	ϕ (°)
F1	0.25	1.0625	0.97014	−0.2633	−14.04
F2	0.50	1.2500	0.89443	−0.9691	−26.57
F3	1.00	2.0000	0.70711	−3.0103	−45.00
F4	1.50	3.2500	0.55470	−5.1188	−56.31
F5	2.50	7.2500	0.37139	−8.6034	−68.20

4.4 Measured gain

Using Equations 4 and 5, with V_{in} and V_{out} read from the oscilloscope. For F1:

$$A_v = 2.42\ \text{V} / 2.50\ \text{V} = 0.968 \tag{15}$$

$$A_v(\text{dB}) = 20 \log_{10}(0.968) = -0.28\ \text{dB} \tag{16}$$

Table 4: Measured gain from oscilloscope readings of V_{in} and V_{out}.

Source	V_{in} (V)	V_{out} (V)	$A_v = V_{out}/V_{in}$	A_v (dB)
F1	2.50	2.42	0.968	−0.28
F2	2.50	2.23	0.892	−0.99
F3	2.50	1.77	0.708	−3.00
F4	2.50	1.39	0.556	−5.10
F5	2.50	0.93	0.372	−8.59

4.5 Percentage difference and table of results

Using Equation 9, for F1:

$$\%\ \text{difference} = (0.968 - 0.970)/0.970 \times 100 = -0.21\ \% \tag{17}$$

Table 5 combines the measured and theoretical results for all five sources.

Table 5: Table of results.

Source	Mult.	f (Hz)	V_{in} (V)	V_{out} (V)	Meas. gain	Meas. gain (dB)	Theor. gain	Theor. gain (dB)	Expected V_{out} (V)	% diff. in gain
F1	0.25	39.78	2.50	2.42	0.968	−0.28	0.970	−0.26	2.425	−0.21
F2	0.50	79.58	2.50	2.23	0.892	−0.99	0.894	−0.97	2.235	−0.22
F3	1.00	159.15	2.50	1.77	0.708	−3.00	0.707	−3.01	1.768	+0.14
F4	1.50	238.73	2.50	1.39	0.556	−5.10	0.555	−5.12	1.387	+0.18
F5	2.50	397.88	2.50	0.93	0.372	−8.59	0.371	−8.60	0.928	+0.27

5. Interpretation of Data

5.1 SimulIDE circuit design

Figure 1 shows the circuit as built in SimulIDE: the five wave generators, the Arduino-driven relay selector, the RC network, the op-amp buffer, and the oscilloscope on V_{in} and V_{out}.

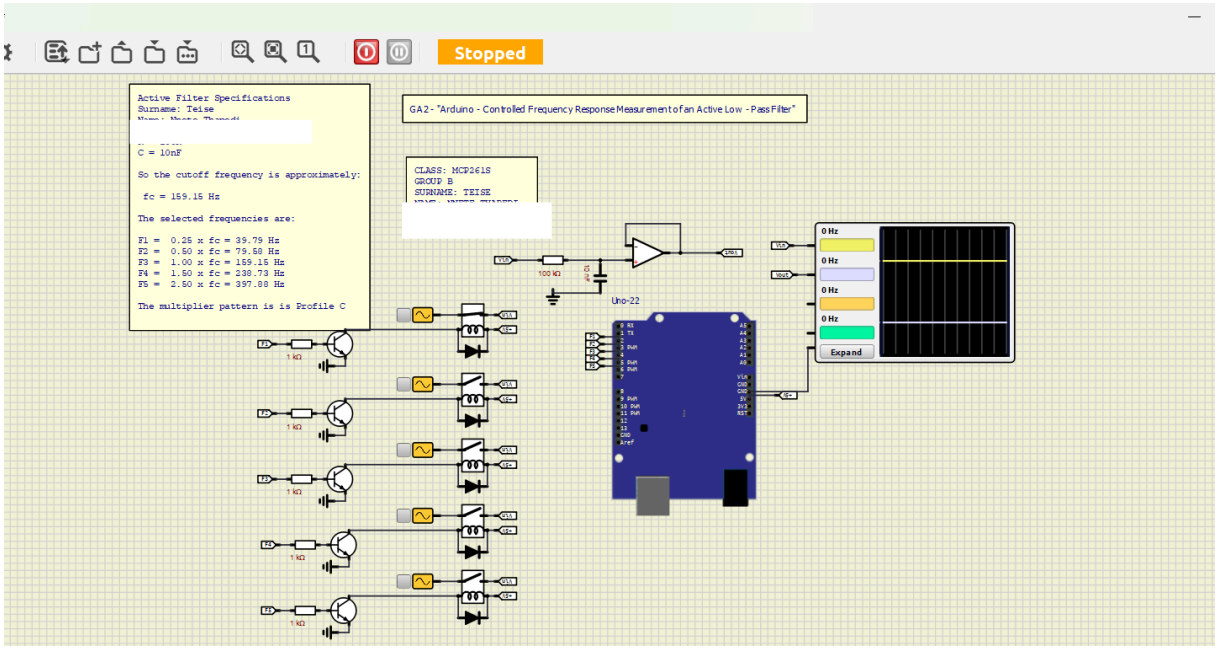


Figure 1: SimulIDE circuit of the Arduino-controlled relay selector, RC network and op-amp buffer.

5.2 Oscilloscope evidence

Figures 2 to 6 show the oscilloscope capture for each source, with V_{in} on Ch1 and V_{out} on Ch2. The input amplitude stays the same from F1 to F5 while the output gets smaller.

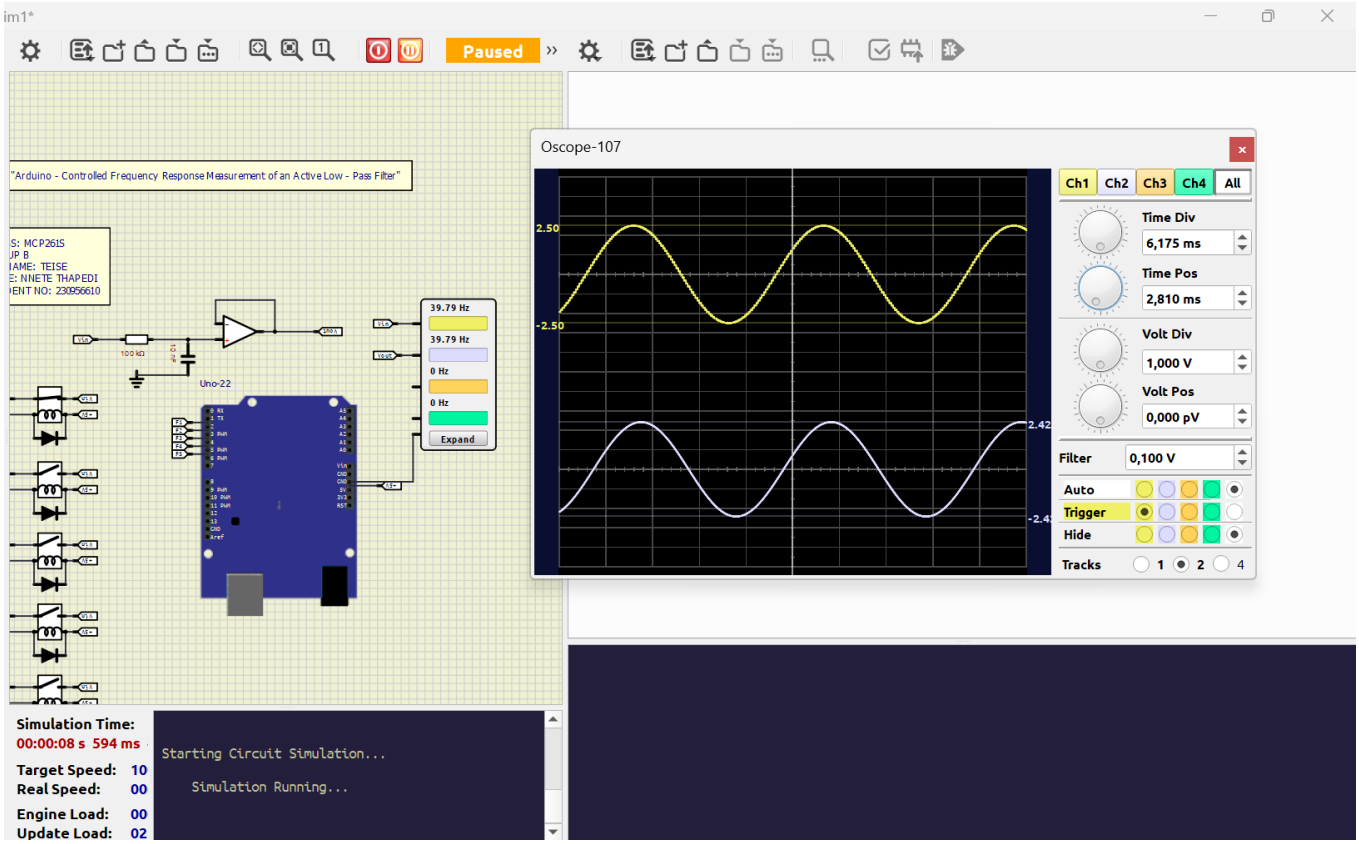


Figure 2: Oscilloscope capture for source F1 (39.78 Hz) (Ch1: V_{in}; Ch2: V_{out}).

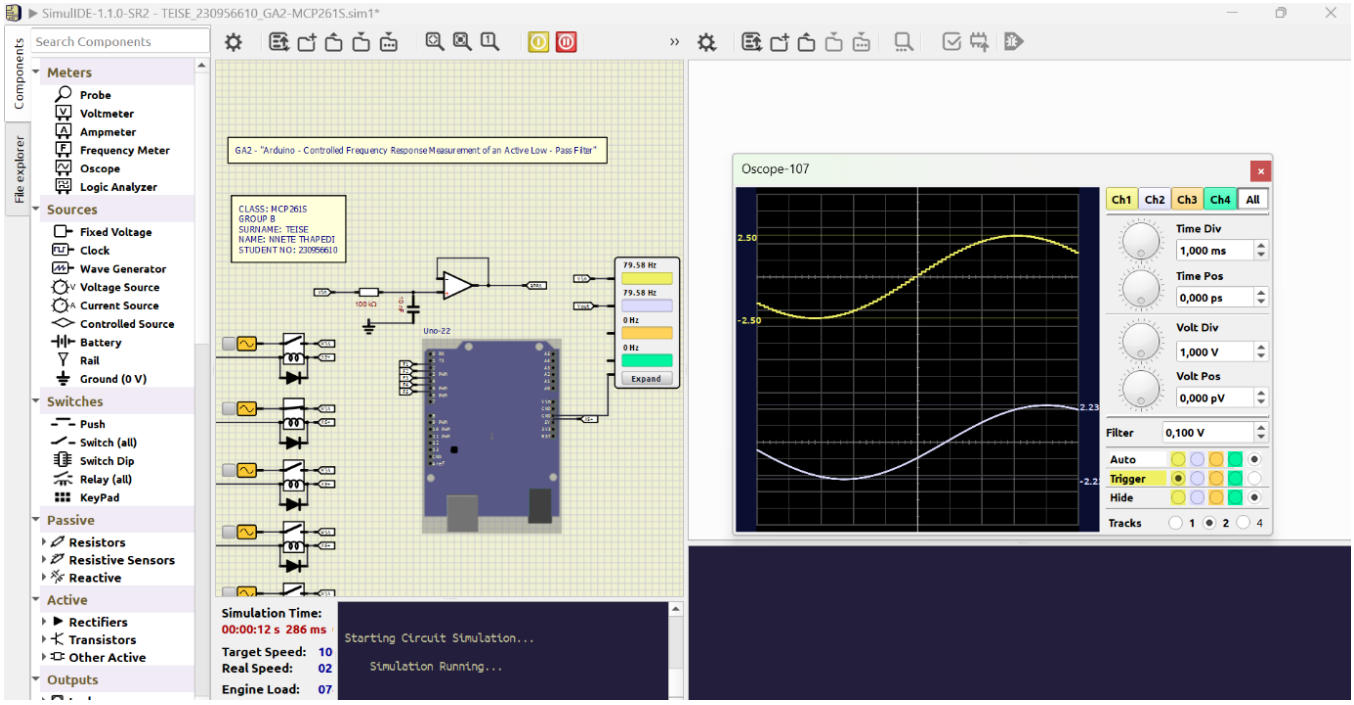


Figure 3: Oscilloscope capture for source F2 (79.58 Hz) (Ch1: V_{in}; Ch2: V_{out}).

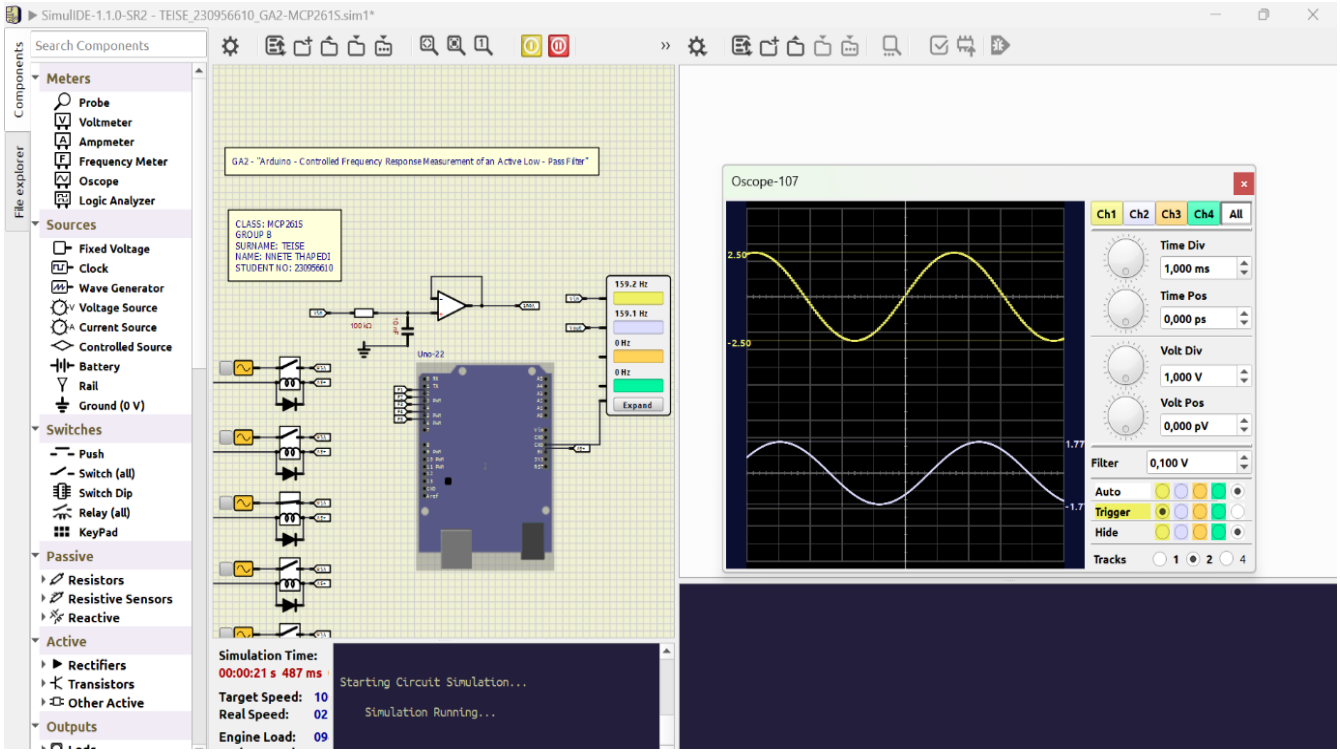


Figure 4: Oscilloscope capture for source F3 (159.15 Hz) (Ch1: V_{in}; Ch2: V_{out}).

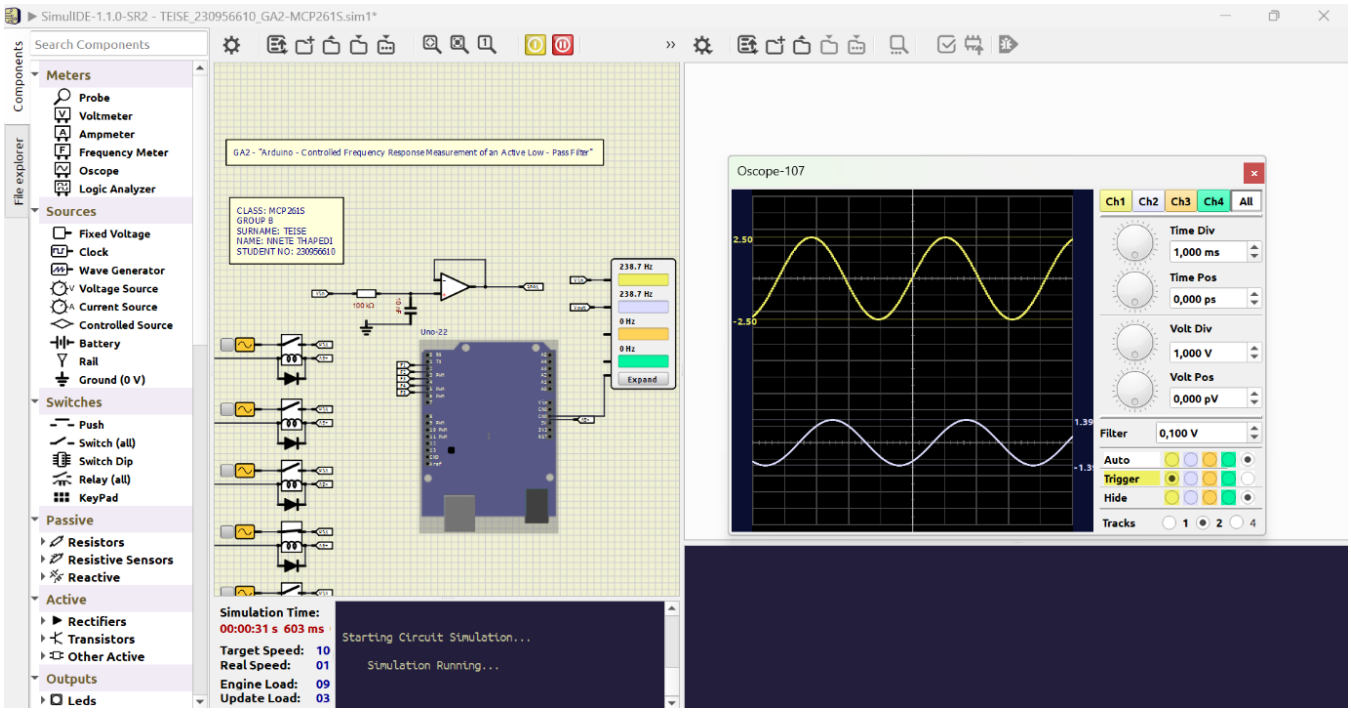


Figure 5: Oscilloscope capture for source F4 (238.73 Hz) (Ch1: V_{in}; Ch2: V_{out}).

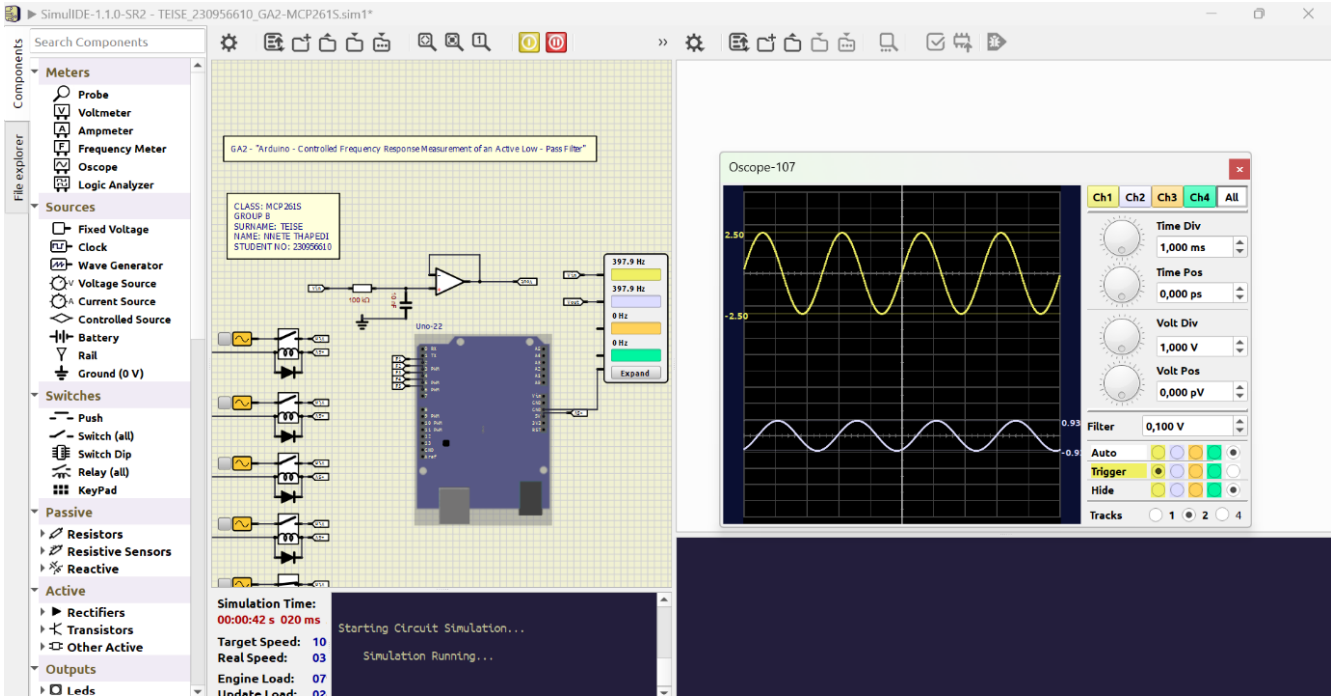


Figure 6: Oscilloscope capture for source F5 (397.88 Hz) (Ch1: V_{in}; Ch2: V_{out}).

5.3 Frequency-response graph

Figure 7 plots measured and theoretical gain in dB against frequency on a logarithmic axis. The passband level is 0 dB because K = 1. The smooth curve is calculated from Equation 6 and is not extra measured data. The −3 dB point is at the cutoff frequency, f_c = 159.155 Hz.

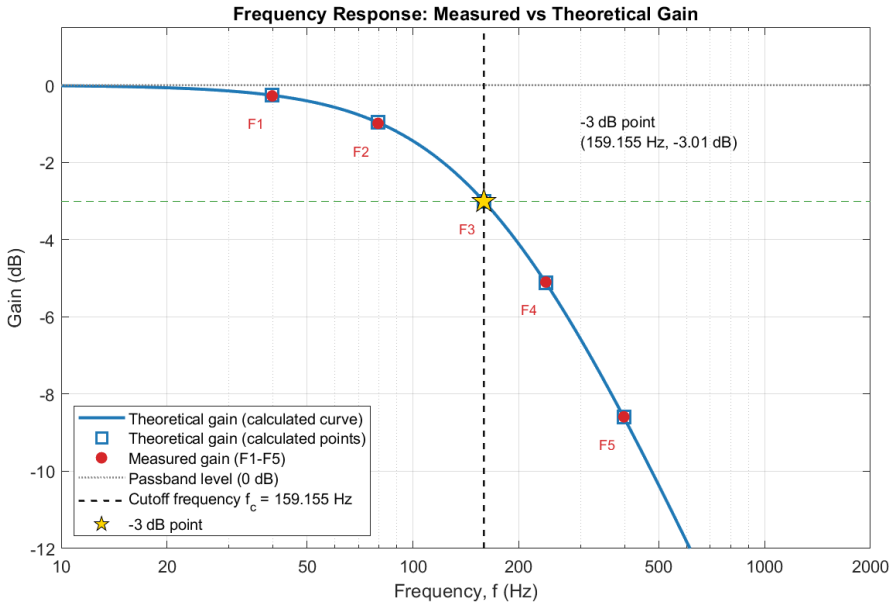


Figure 7: Measured and theoretical gain (dB) against frequency (Hz) on a logarithmic frequency axis. The theoretical curve is calculated.

5.4 Trend from low to high frequency

Measured gain falls steadily as frequency rises: 0.968 (−0.28 dB) at F1 = 39.78 Hz, 0.892 (−0.99 dB) at F2 = 79.58 Hz, 0.708 (−3.00 dB) at F3 = 159.15 Hz, 0.556 (−5.10 dB) at F4 = 238.73 Hz and 0.372 (−8.59 dB) at F5 = 397.88 Hz. This is the pattern expected from a first-order low-pass filter (Sedra et al., 2020). Only F4 and F5 are above the cutoff frequency, which is not enough to measure the roll-off rate the filter approaches well above f_c.

5.5 Cutoff region

For R = 100 kΩ and C = 10 nF the calculated cutoff frequency is 159.155 Hz. At F3, V_{in} = 2.50 V and V_{out} = 1.77 V, so the measured gain is 0.708 (−3.00 dB). Theory gives 0.707 (−3.01 dB), a difference of about 0.01 dB. The theoretical phase shift at cutoff is −45°, a time lag of about 0.785 ms. Phase was not measured, so this is a prediction and not a result.

5.6 Agreement between theory and simulation

Measured and theoretical gains are 0.968/0.970 (F1), 0.892/0.894 (F2), 0.708/0.707 (F3), 0.556/0.555 (F4) and 0.372/0.371 (F5). The percentage differences are −0.21%, −0.22%, +0.14%, +0.18% and +0.27%. All are below 0.3% in size, and the error does not grow with frequency. The sign also changes across the range, so there is no sign of a consistent gain error.

5.7 Possible sources of difference

- The oscilloscope grid can only be read to limited precision, and voltages were recorded to two decimal places.
- The percentage differences in Table 5 come from gains rounded to three decimal places. With unrounded theoretical gains they become −0.22%, −0.27%, +0.13%, +0.23% and +0.16%, which is at most 0.11 percentage points different and still under 0.3%. The expected output voltages were also worked out from rounded gains, so F2 would be 2.236 V rather than 2.235 V.
- The generator frequencies in Table 5 are given to two decimal places and are within 0.01 Hz of the exact values in Table 2, which has a negligible effect on gain.
- Each source is connected for 10 s after a 0.3 s all-off gap. A reading taken before the waveform has settled after a relay closes could be off. The screenshots do not show the settling time of every waveform.
- Only one source is connected at a time because of the break-before-make switching, and V_{in} was measured after the relay selector and not taken from the generator setting.
- The op-amp operating range matters because the supply rails must exceed the 2.50 V input peak. Clipping occurred only in the earlier test with too little negative supply (Section 2.4).

5.8 Unexpected results

The five gains fall steadily with frequency, and none stands out as an outlier against theory. There was no gain reversal or sudden jump. The only unexpected behaviour was the clipped negative half of V_{out} in the earlier test with too little negative supply (Section 2.4), which went away once a dual supply was used.

5.9 Limitations

- SimulIDE is a simulation, not a hardware test, so component tolerances, op-amp bandwidth effects and noise are not examined.
- Only five frequencies were measured, so the smooth line in Figure 7 is theoretical.
- The input amplitude is 2.50 V in every row of Table 5, and the zero-centred waveforms in Figures 2 to 6 fit a peak-amplitude convention. The generator property readouts and DC offsets are not shown separately.
- The op-amp rail voltages were not recorded, so no exact values are given.
- No phase measurement was made, so the −45° at cutoff is a prediction only.

5.10 Conclusion

Across the five test frequencies the simulated active low-pass filter reduced the measured gain from 0.968 (−0.28 dB) at 39.78 Hz to 0.372 (−8.59 dB) at 397.88 Hz. At the calculated cutoff frequency of 159.155 Hz the measured gain was 0.708 (−3.00 dB), very close to the expected 0.707 (−3.01 dB). These results support the first-order, unity-passband-gain model, with differences from theory of up to 0.27% as shown in Table 5.

The Arduino-controlled relay selector connected one source at a time using transistor drivers and flyback diodes, and the op-amp buffer kept the measurement load from affecting the RC network. The results do not by themselves show how a physical build would perform, and they do not confirm a measured −45° phase shift. Building the circuit in hardware, with recorded rail voltages and a phase measurement, would be the natural next step.

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Appendix A: Handwritten Calculations

The handwritten workings behind the typed calculations in Section 4 are shown in Figures A1 to A5. The expected output voltages in these sheets were worked out from the three-decimal-place gains.

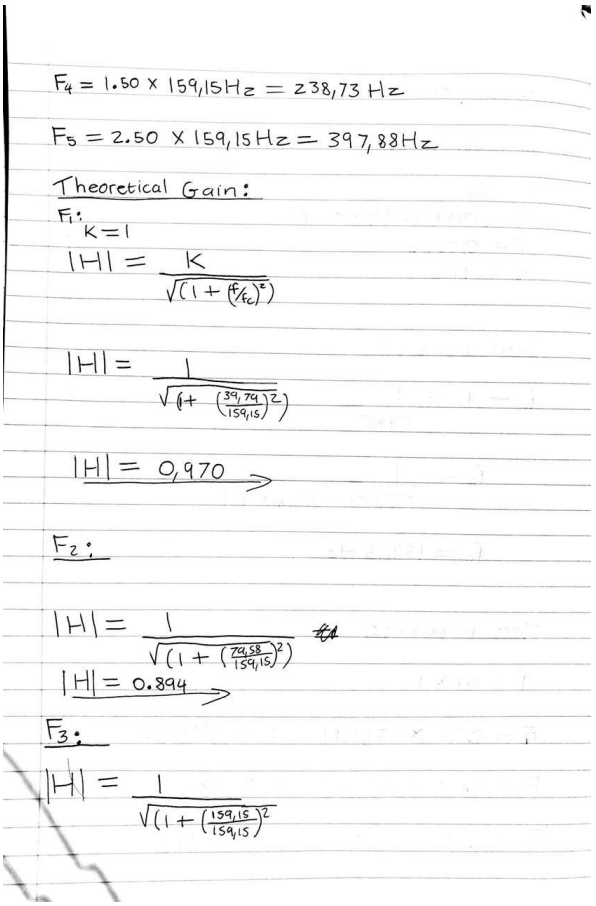


Figure A1: Test frequencies F4 and F5; theoretical gain |H| for F1 to F3.

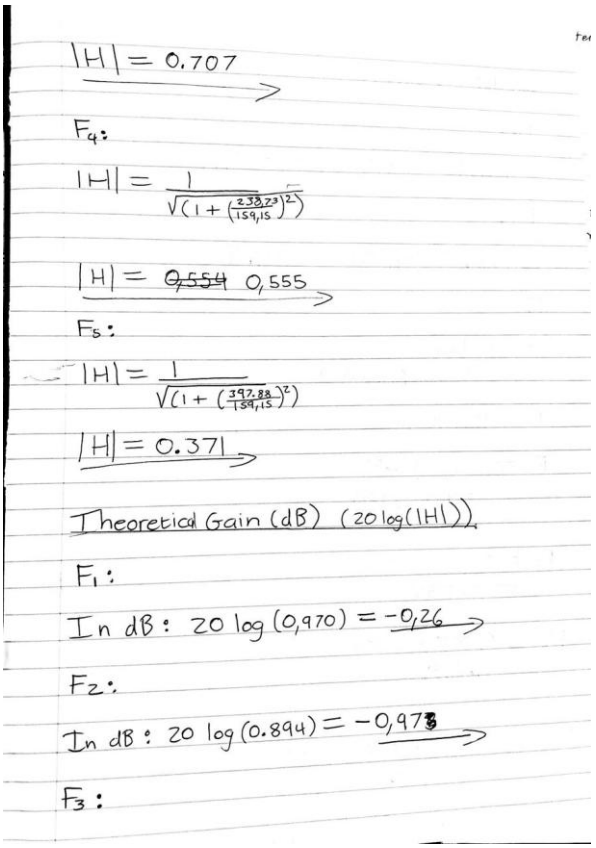


Figure A2: Theoretical gain |H| for F3 to F5; theoretical gain in dB for F1 to F3.

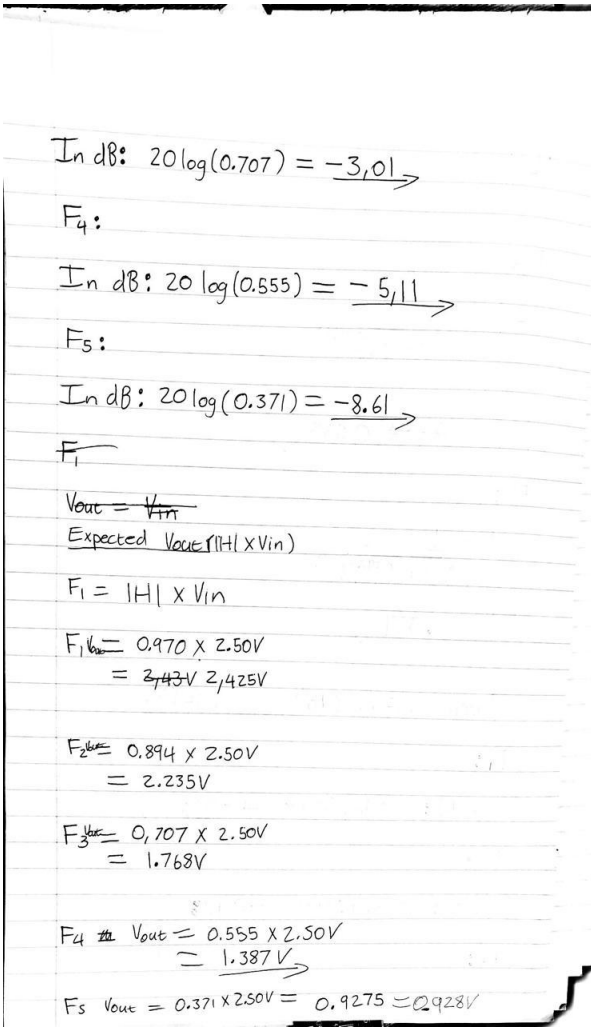


Figure A3: Theoretical gain in dB for F3 to F5; expected V_{out}.

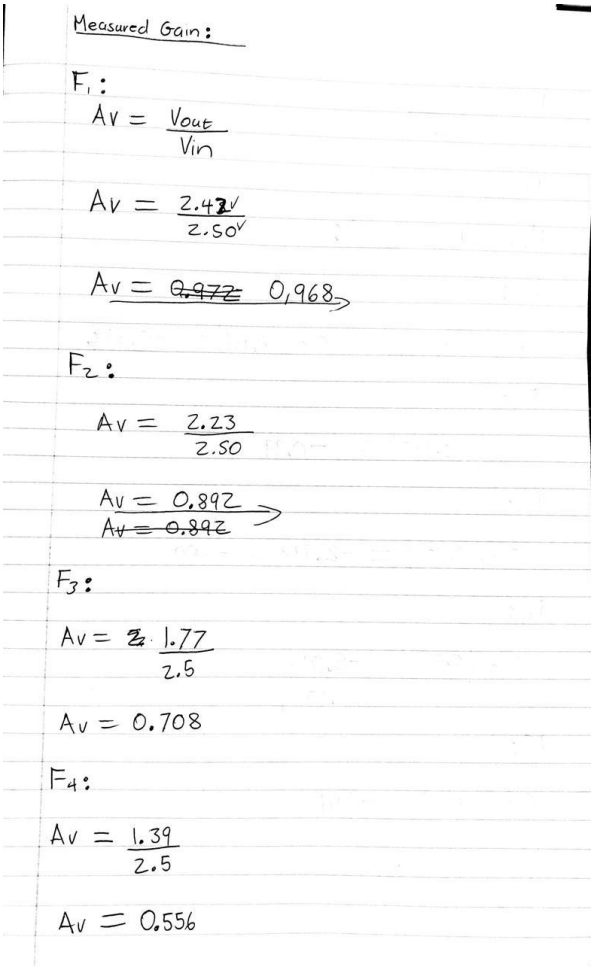


Figure A4: Measured gain for F1 to F4.

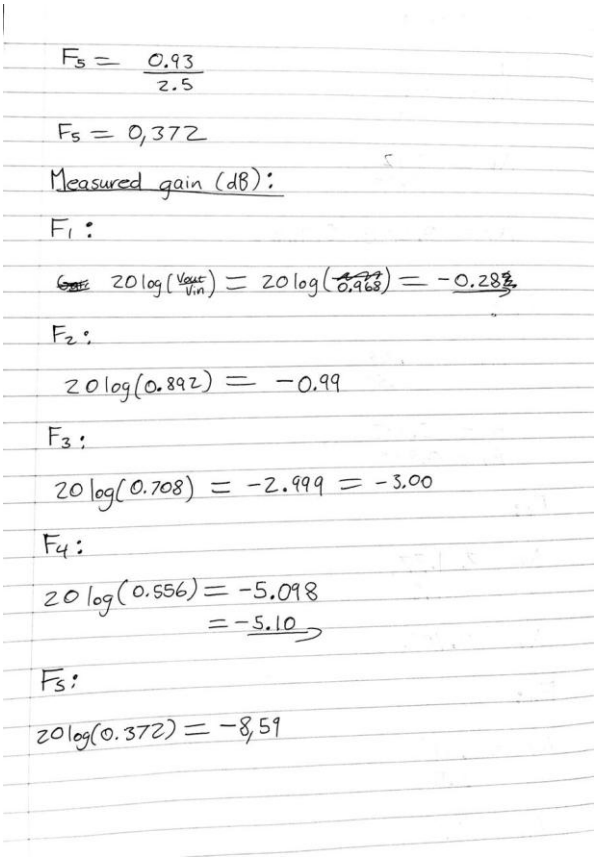


Figure A5: Measured gain for F5; measured gain in dB.